TriFrame: can pictures of stock charts help predict risk?

Computer Vision with Deep Learning project · end-to-end report · 500-stock run (s500, notebook v4) · generated 2026-10-21

In one paragraph. We asked whether pictures of a stock's price chart at three time scales (weekly, daily and hourly) help a deep-learning model predict three things about the coming weeks: how much the price will swing (**volatility**), whether it will fall 10% or more at some point (**drawdown**), and whether it will beat the typical stock over the next 5 days (**direction**). We tested 7 pretrained vision models (each both frozen and fine-tuned) on 500 US stocks (236,520 chart sets, 2001-2022), always against a simple baseline that uses only 33 numbers computed from the same prices. **The answer:** risk is predictable (volatility R^2 0.50 on the 2020-2022 test years (a historical check: earlier runs had already looked at them)), but the chart pictures did not add a significant improvement over the numbers there, and there is no reliable 5-day direction signal, so the system's default decision is **NO TRADE**.

Target	Model chosen	Test score (2020-2022)	2023-2026 (already viewed)	In plain words
Volatility, next 20 days	ConvNeXt-Tiny (fine-tuned), pretrained on ImageNet-12k + 1k + gradient-boosted trees (XGBoost)	R^2 0.50	not opened	explains about 50% of the variation in volatility
10%+ drawdown within 20 days	ConvNeXt-Tiny (fine-tuned), pretrained on ImageNet-12k + 1k + gradient-boosted trees (XGBoost)	AUC 0.65	not opened	ranks a stock that drops 10% above one that does not 65% of the time
Beats the median stock, 5 days	numbers only + gradient-boosted trees (XGBoost)	AUC 0.510	not opened	no better than a coin flip

What is new in this version (v4)

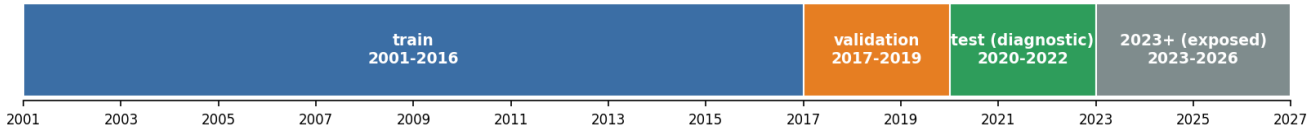
- Trained on **500 stocks** (the sample contains the 60 and 120 stocks of the earlier runs, so all three compare on the same rows).
- **Did training reach the minimum?** Certified global minima for the convex parts, minimum-of-validation-loss and loss-landscape evidence for the networks (a global minimum cannot be certified for a neural network).
- **Training, validation and loss curves** for every fine-tuned network, the boosted trees and the regularised heads.
- **Precision, recall, F1, ROC-AUC, R^2 and RMSE** with ROC and precision-recall curves, confusion matrices and reliability plots.
- **Fixes from an external review**: fine-tuning no longer uses the selection years, volatility baselines (EWMA, GARCH), corrected metrics, and a provenance record for every number.

Honest status of the data used for testing. The 2020-2022 test years were already read by the earlier runs, so here they are a historical check, not a fresh test; the 2023-2026 period has been opened 2 times before, so it is no longer an untouched holdout either. Every choice in this run was made on the 2017-2019 validation years only. A truly new test needs data after October 2026.

Steps 1-2: the data and how it was split

Each example, a **chart set**, is one stock on one day: three black-and-white chart pictures drawn from its price history (104 weekly bars, 126 daily bars and 140 hourly bars) plus 33 ordinary numbers (returns, price ranges, volatility measures) computed from the last 126 daily bars. The labels are what happened next: the realised volatility over the next 20 trading days, whether the price traded 10% below the next day's open within 20 days, and whether the stock beat the median stock over 5 days. This run uses **500 of 979 US stocks** (chosen at random, and including the 60 stocks of the earlier pilot), 236,520 chart sets in total.

Time is never shuffled. Models learn from the **training** years, are tuned and selected on the **validation** years, and are then checked on the **test** years (already seen by earlier runs, so a historical check). 2023 onward was a sealed **holdout** but has been opened before, so this run uses it for nothing. A 20-day gap at each boundary stops any label from leaking into the next period. Rows are stock-days with overlapping 20-day windows, so they are not independent: every confidence interval resamples whole calendar months.



Period	Years	Chart sets	Stocks	Used for
train	2001-2016	165,421	489	fit the models
validation	2017-2019	35,145	492	tune, select, calibrate, set thresholds and the simulation
test	2020-2022	35,954	500	historical diagnostic: already read by earlier runs
2023+	2023-2026	-	-	exposed: opened 2 times before



What a model sees: the three pictures of one chart set (drawn here from synthetic prices). Each picture is scaled to fill its panel, so it shows the shape of the moves, not their size in dollars.

Step 3: the pretrained vision models

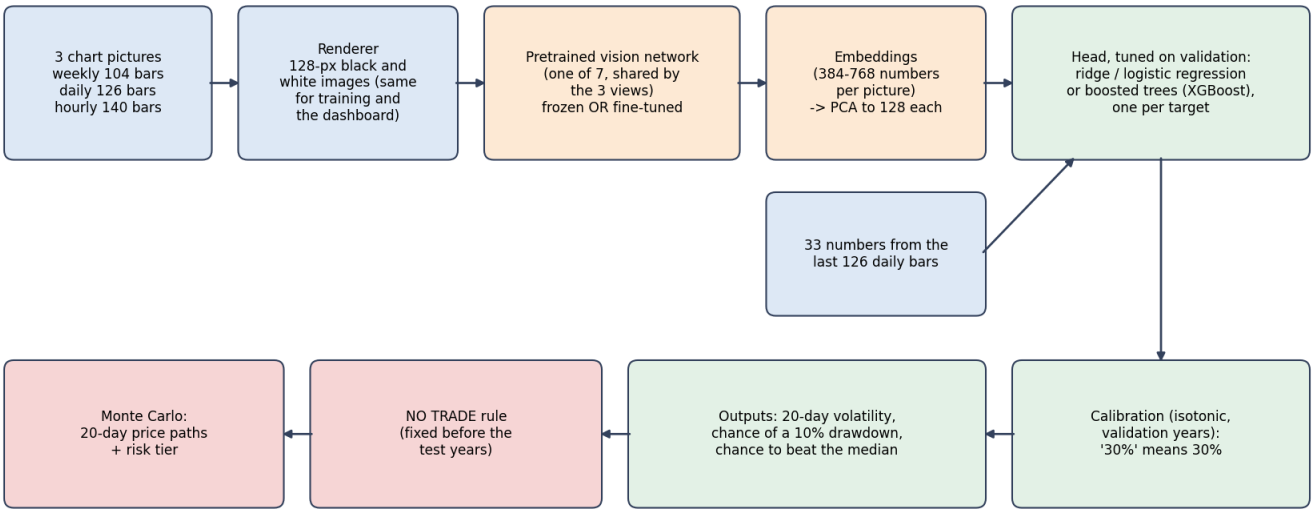
A **pretrained** model is a neural network that has already learned to recognise edges, shapes and textures from millions of ordinary photos. We reuse that knowledge instead of training a network from scratch. We compared seven well-known designs: convolutional networks (CNNs), which scan the picture with small filters, and transformers, which split it into patches that 'pay attention' to each other. Three ResNet-18s with **random, untrained weights** serve as a control: if they did as well, pretraining would not be doing anything.

Model	Type	Pretrained on	Parameters (millions)	Fine-tuning
ResNet-18	CNN	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	11.2	4 epochs (best 2), 4 min
EfficientNet-B0	CNN	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	4.0	4 epochs (best 2), 11 min
ViT-B/16	Transformer	ImageNet-21k, then fine-tuned on ImageNet-1k	85.8	3 epochs (best 1), 10 min

Model	Type	Pretrained on	Parameters (millions)	Fine-tuning
ConvNeXt-Tiny	CNN	ImageNet-12k, then fine-tuned on ImageNet-1k	27.8	4 epochs (best 2), 11 min
Swin-T	Transformer	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	27.5	5 epochs (best 3), 16 min
DINOv2 ViT-S/14	Transformer	LVD-142M, 142 million unlabelled images (self-supervised)	21.6	7 epochs (best 5), 15 min
MobileNetV3-Large	CNN	ImageNet-1k (1.28 million labelled photos, 1,000 classes)	4.2	5 epochs (best 3), 11 min
ResNet-18 (random weights, seed 0)	Control	none (random weights)	11.2	-
ResNet-18 (random weights, seed 1)	Control	none (random weights)	11.2	-
ResNet-18 (random weights, seed 2)	Control	none (random weights)	11.2	-

Frozen or fine-tuned? 'Frozen' means the network's own weights are left exactly as they came from pretraining; we only read its output (an **embedding**, a list of a few hundred numbers summarising the picture) and train small models on top. 'Fine-tuned' means the network's weights are also adjusted on our charts. **In this run every pretrained network was fine-tuned** (early stopping on 2015-2016, inside the training years), plus a controlled study and extra seeds.

Step 4: the architecture



- The renderer draws the three charts as 128-pixel-high black-and-white pictures; the dashboard uses the very same code.
- One vision network turns each picture into an embedding; **PCA** compresses each embedding to 128 numbers (fitted on training years only).
- The three compressed embeddings and the 33 numbers go into a **head**: ridge or logistic regression, or gradient-boosted decision trees (XGBoost). One head per target.
- Probabilities are **calibrated** on the validation years so that '30%' really means 30%.
- A **NO TRADE rule**, fixed before the test years were opened, turns the outputs into a decision, and a **Monte Carlo simulation** turns the volatility forecast into a fan of possible price paths.

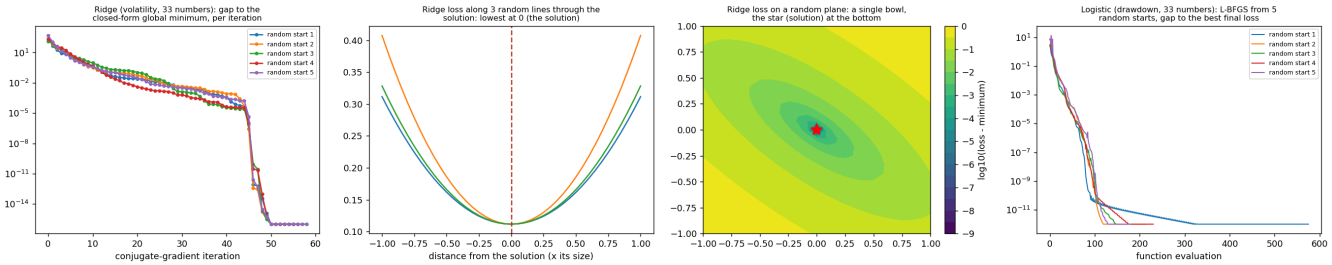
Step 5: training and hyper-parameter tuning

We tried **1,881 model configurations** and logged every one: ridge strengths, logistic regularisation, tree depth, learning rate and the number of trees (by early stopping), for each target, each vision network and each combination of inputs (images only, numbers + all three timeframes, numbers + one timeframe). Every choice was made on the validation years only, and all preprocessing (gap filling, scaling, PCA) was fitted on the training years only. **The selection rule was written down before the test years were opened:** per target the best validation score wins, and a model that uses chart pictures wins only if its improvement over the numbers-only model is clearly above zero (its 95% confidence interval excludes 0).

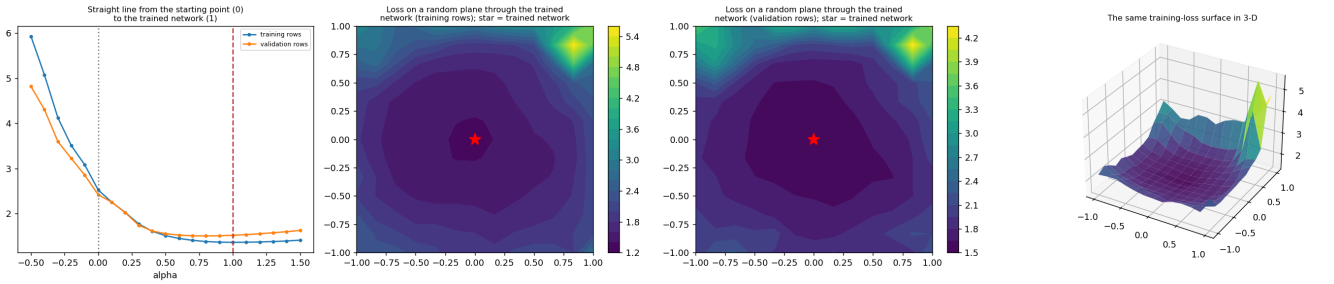
Did the models reach the global minimum?

Training a model means searching for the settings with the lowest loss (error). For two kinds of model the answer can be proven: the **ridge** and **logistic** heads are **convex**, which means their loss surface is a single bowl with exactly one lowest point, so the minimum found is the global minimum. We certify it three ways: the slope (gradient) there is zero, every random starting point ends at the same answer, and the loss rises in every direction around it. **Boosted trees** are built step by step; their validation loss has a lowest point, which early stopping keeps. **Neural networks are not convex**: no method can prove a global minimum for them, and the lowest training loss is not even the goal, because it usually means memorising the training years. What we show instead is that training converged, the kept checkpoint is the lowest point of the held-out loss, the trained network sits at the bottom of its valley in the loss landscape, and different random seeds land in the same place.

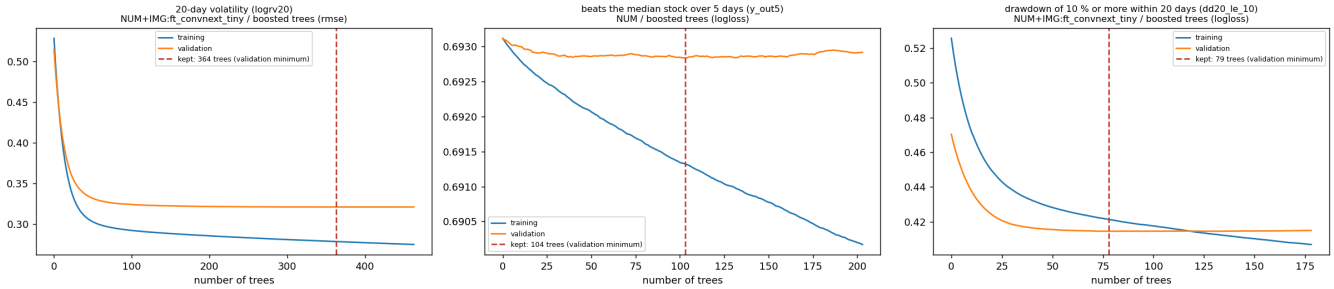
Part of the model	Convex?	Verdict
Ridge head (volatility)	yes	GLOBAL minimum reached (certified)
Logistic head (drawdown)	yes	GLOBAL minimum reached (certified)
Boosted trees (V, NUM+IMG:ft_convnext_tiny)	no (greedy)	minimum of the validation curve kept (early stopping)
Boosted trees (O5, NUM)	no (greedy)	minimum of the validation curve kept (early stopping)
Boosted trees (D10, NUM+IMG:ft_convnext_tiny)	no (greedy)	minimum of the validation curve kept (early stopping)
Fine-tuned network (ConvNeXt-Tiny (fine-tuned))	no	converged to a stable minimum of the held-out loss; a GLOBAL minimum cannot be certified for a network



Left to right: ridge, gap to the closed-form minimum from 5 random starts; ridge loss along random lines and on a random plane (lowest at the solution); logistic regression from 5 random starts.



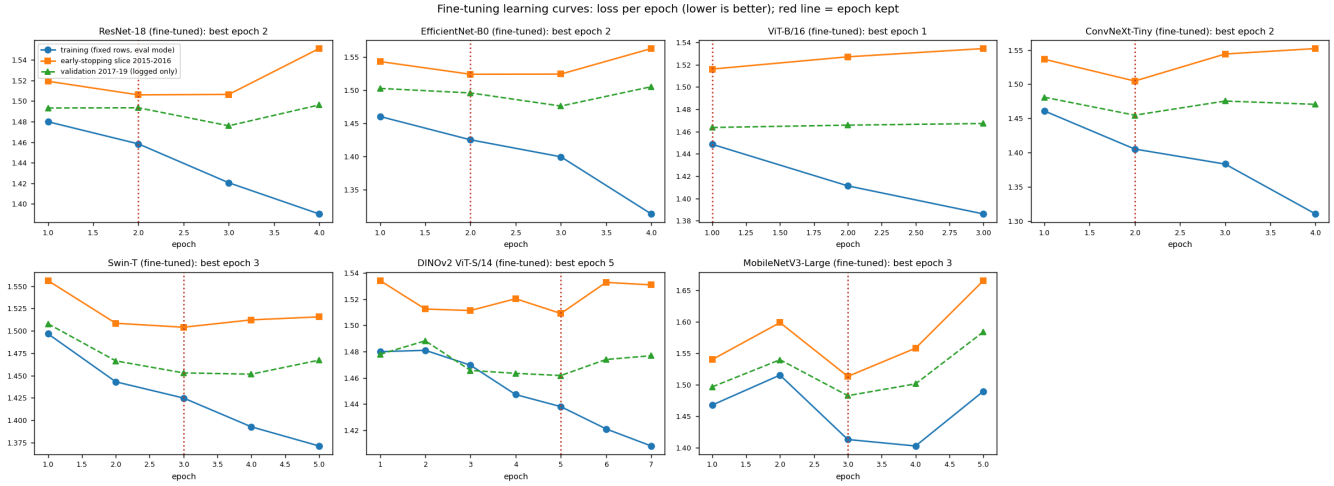
The fine-tuned network's loss landscape: along the straight line from its starting point (0) to the trained network (1), and on random planes through it (star = trained network).

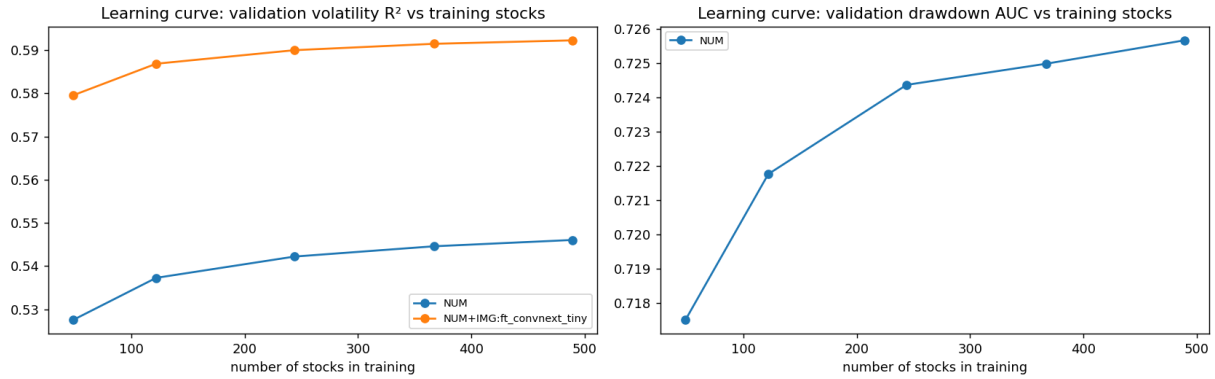


Boosted trees: training and validation loss as trees are added; the dashed line is the validation minimum that was kept.

Training, validation and loss curves

Each panel follows one network through fine-tuning: the loss on fixed training rows (eval mode), on the early-stopping slice inside the training years, and on the validation years (logged only). Training loss keeps falling while the held-out losses turn up after a few epochs: that is overfitting, and the epoch with the lowest early-stopping loss (red line) is the one kept.

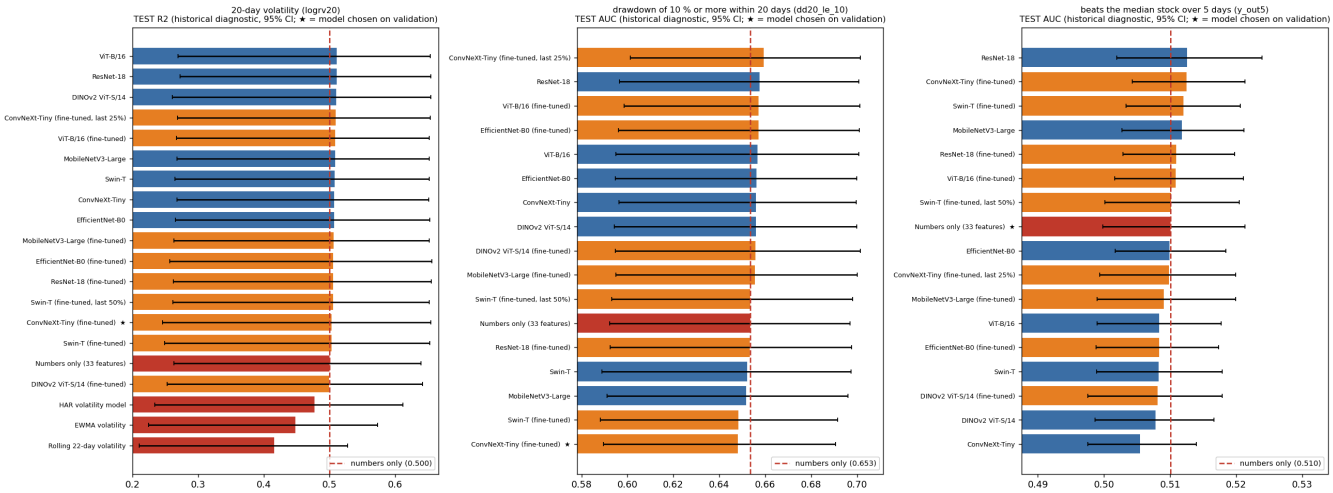




Learning curves: validation score as the number of training stocks grows (does more data help?).

Step 6: results - which models won, and how good they are

Target	Inputs	Vision network	Head	Validation	Test (95% CI)
Volatility, next 20 days	33 numbers + weekly + daily + hourly charts	ConvNeXt-Tiny (fine-tuned), pretrained on ImageNet-12k + 1k	gradient-boosted trees (XGBoost)	R^2 0.592	R^2 0.503 (0.25 to 0.65)
10%+ drawdown within 20 days	33 numbers + weekly + daily + hourly charts	ConvNeXt-Tiny (fine-tuned), pretrained on ImageNet-12k + 1k	gradient-boosted trees (XGBoost)	AUC 0.739	AUC 0.648 (0.59 to 0.69)
Beats the median stock, 5 days	33 numbers from the last 126 daily bars	none (numbers only)	gradient-boosted trees (XGBoost)	AUC 0.514	AUC 0.510 (0.500 to 0.521)



Every candidate on the test years (historical diagnostic; 95% CIs; ★ = the model chosen on validation).

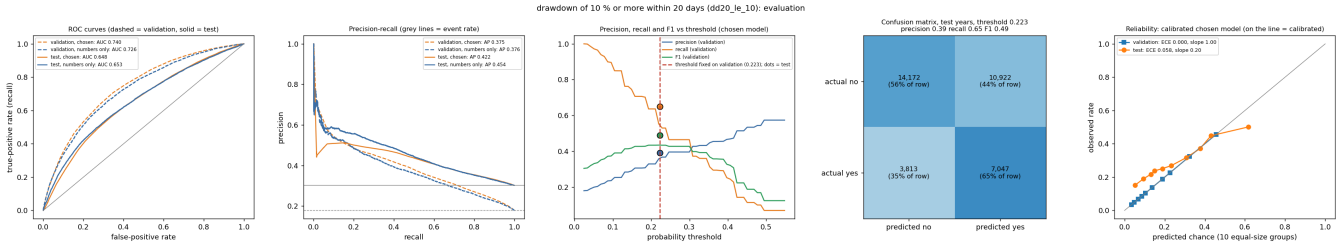
- **Do chart pictures add to the numbers? No.** On the test years, adding all three chart views to the numbers changed volatility R^2 by +0.003 (95% CI -0.018 to +0.026) and direction AUC by -0.002 (95% CI -0.009 to +0.004). Both intervals include zero.
- **Do the vision models differ from each other? Yes, a little.** 10 of 21 pairs of frozen pretrained networks differ in how well their pictures alone predict volatility (validation, Holm-corrected); the best reaches R^2 0.207.
- **Does waiting for confident direction calls help? No.** Keeping only the 10% most confident direction calls changed their accuracy by +0.008 (95% CI -0.005 to +0.019) on the test years.

- **Sanity checks.** With shuffled labels the pipeline finds nothing (volatility R² -0.123, direction AUC 0.495), and randomly initialised ResNet-18s reach volatility R² -0.021 to -0.017 from pictures alone, against 0.207 for the best pretrained network.

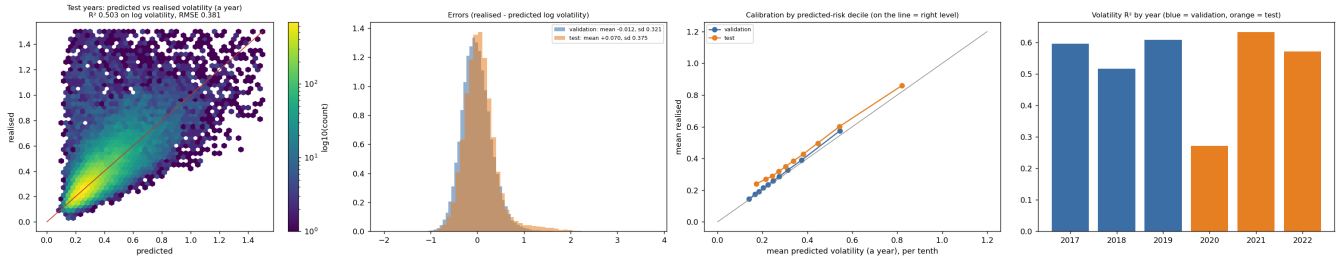
Evaluation metrics: precision, recall, F1, ROC-AUC, R² and RMSE

R² is the share of the ups and downs explained (volatility); **RMSE** the typical error in log volatility. For the yes/no targets, a stock is flagged when its predicted chance passes a threshold fixed on the validation years (the one with the best F1: drawdown 0.22, direction 0.44). **Precision** is the share of flagged stocks that really had the event; **recall** the share of events that were flagged; **F1** balances the two; **ROC-AUC** is the chance a stock with the event is ranked above one without. Accuracy alone would mislead: if 29% of cases have a drawdown, always saying 'no' is already 71% accurate.

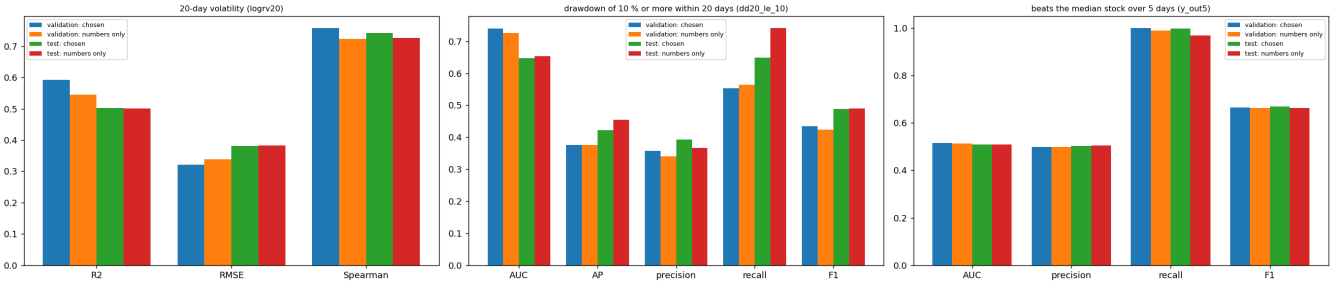
Target	Years	Model	Score			
20-day volatility	validation	chosen model	R² 0.592	RMSE 0.321	-	-
20-day volatility	validation	numbers only	R² 0.546	RMSE 0.338	-	-
20-day volatility	test	chosen model	R² 0.503	RMSE 0.381	-	-
20-day volatility	test	numbers only	R² 0.500	RMSE 0.382	-	-
drawdown of 10 % or more within 20 days	validation	chosen model	AUC 0.740	precision 0.36	recall 0.55	F1 0.43
drawdown of 10 % or more within 20 days	validation	numbers only	AUC 0.726	precision 0.34	recall 0.56	F1 0.42
drawdown of 10 % or more within 20 days	test	chosen model	AUC 0.648	precision 0.39	recall 0.65	F1 0.49
drawdown of 10 % or more within 20 days	test	numbers only	AUC 0.653	precision 0.37	recall 0.74	F1 0.49
beats the median stock over 5 days	validation	chosen model	AUC 0.516	precision 0.50	recall 1.00	F1 0.67
beats the median stock over 5 days	validation	numbers only	AUC 0.514	precision 0.50	recall 0.99	F1 0.66
beats the median stock over 5 days	test	chosen model	AUC 0.510	precision 0.50	recall 1.00	F1 0.67
beats the median stock over 5 days	test	numbers only	AUC 0.510	precision 0.50	recall 0.97	F1 0.66



Drawdown model: ROC and precision-recall curves, precision / recall / F1 against the threshold, the test-year confusion matrix and the reliability plot.



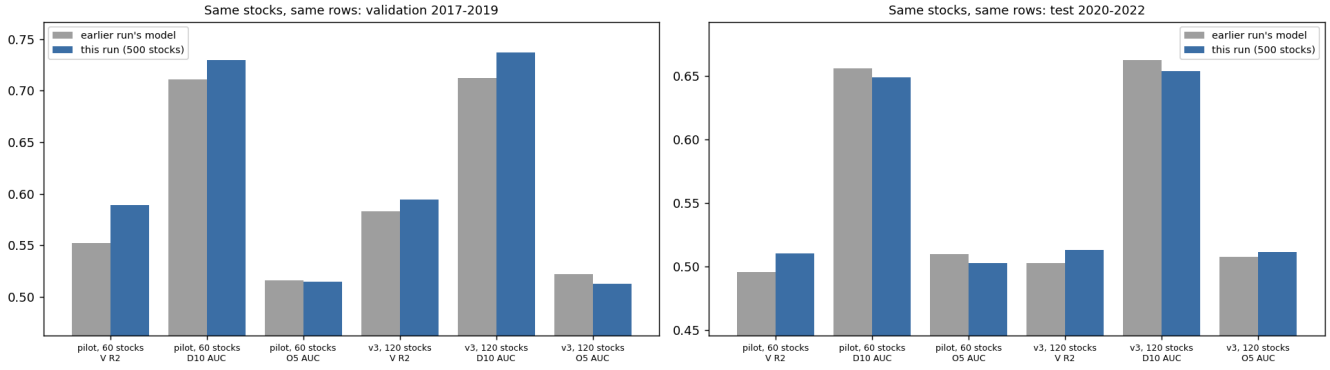
Volatility model: predicted vs realised, errors, calibration by risk decile, and R² by year.



Did R² improve? 60, 120 and this run's stocks on the same rows

Each earlier run's frozen model and this run's model are scored on the earlier run's own stocks and exactly the same rows, so the comparison isolates what changed (mainly more training data). Validation = the selection years; test = the historical check.

Earlier run	Target	Metric	Earlier, validation	This run, validation	Earlier, test	This run, test
pilot, 60 stocks	20-day volatility	R2	0.553	0.589	0.496	0.510
pilot, 60 stocks	drawdown of 10 % or more within 20 days	AUC	0.711	0.730	0.656	0.649
pilot, 60 stocks	beats the median stock over 5 days	AUC	0.516	0.515	0.510	0.503
v3, 120 stocks	20-day volatility	R2	0.583	0.594	0.503	0.513
v3, 120 stocks	drawdown of 10 % or more within 20 days	AUC	0.713	0.737	0.663	0.654
v3, 120 stocks	beats the median stock over 5 days	AUC	0.522	0.513	0.508	0.511



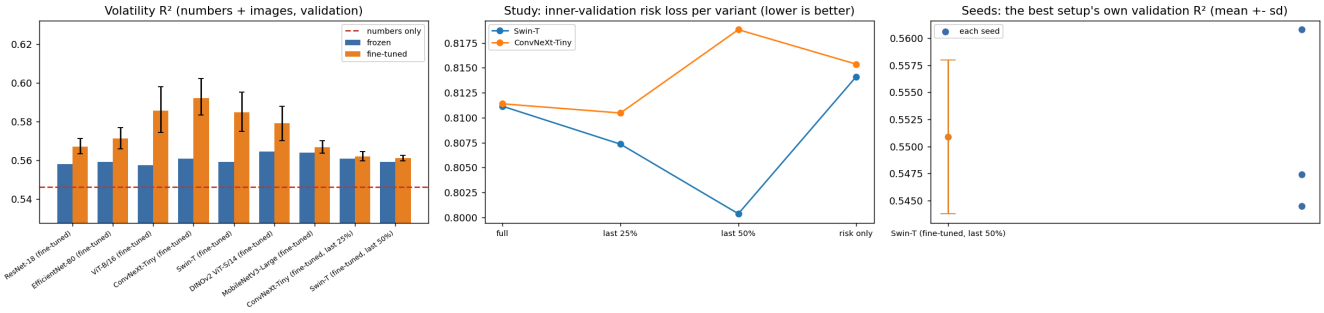
Direction as a ranking (separate experiment). Fitting the continuous 5-day excess return and scoring weekly rankings: NOT SUPPORTED (no reliable cross-sectional ranking signal).

Step 7: did we fine-tune, and did it help?

With early stopping on 2015-2016 (so the 2017-2019 validation years were never used by fine-tuning), fine-tuning raised validation volatility R² (numbers + images) for 9 of 9 runs (significantly for 7, significantly worse for 0; mean change +0.0141) and drawdown AUC for 8 of 9 (significantly for 3). Total fine-tuning time 158 GPU-minutes (15 runs, including the study and the extra seeds).

Model	Frozen R²	Fine-tuned R²	Change (95% CI)	Training time
ResNet-18 (fine-tuned)	0.558	0.567	+0.009 (+0.005 to +0.013)	4 min

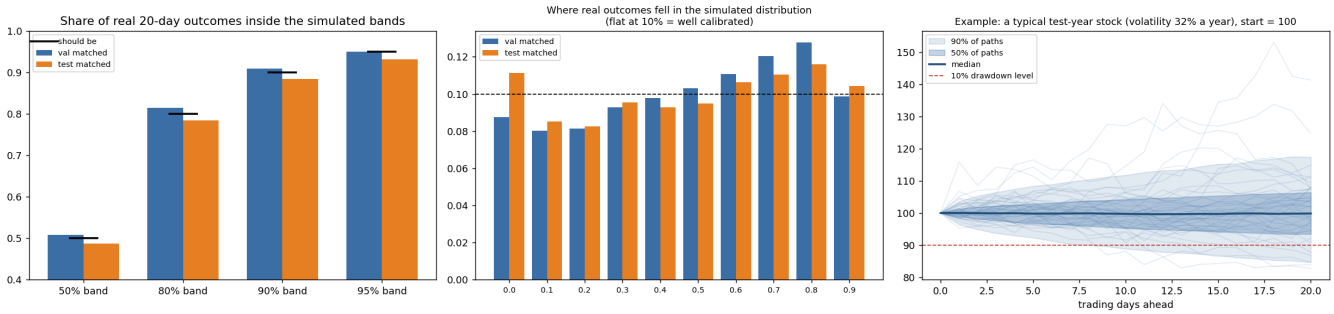
Model	Frozen R²	Fine-tuned R²	Change (95% CI)	Training time
EfficientNet-B0 (fine-tuned)	0.559	0.571	+0.012 (+0.007 to +0.018)	11 min
ViT-B/16 (fine-tuned)	0.558	0.586	+0.028 (+0.017 to +0.041)	10 min
ConvNeXt-Tiny (fine-tuned)	0.561	0.592	+0.032 (+0.023 to +0.042)	11 min
Swin-T (fine-tuned)	0.559	0.585	+0.025 (+0.016 to +0.036)	16 min
DINOv2 ViT-S/14 (fine-tuned)	0.564	0.579	+0.015 (+0.006 to +0.023)	15 min
MobileNetV3-Large (fine-tuned)	0.564	0.567	+0.003 (-0.000 to +0.006)	11 min
ConvNeXt-Tiny (fine-tuned, last 25%)	0.561	0.562	+0.001 (-0.001 to +0.004)	5 min
Swin-T (fine-tuned, last 50%)	0.559	0.561	+0.002 (+0.000 to +0.003)	13 min



Step 8: from predictions to a decision, and the Monte Carlo simulation

NO TRADE rule. A stock is flagged as high risk when its predicted volatility is at least 34% a year or its drawdown chance at least 30% (the top 20% of the model's readings in the validation years). On the test years the flag picked out 48% of cases, and those cases really were riskier: realised volatility 56% vs 30% a year, and 10% drawdowns 40% vs 21% of the time. A direction call would only ever be shown if the direction model had passed its validation *and* test checks; it did not, so every decision is NO TRADE.

Monte Carlo. For each stock the dashboard simulates thousands of 20-day price paths. Each path draws a volatility from the model's forecast and its measured error on validation data, then fat-tailed daily moves (Student-t with 5 degrees of freedom, chosen on validation) with exactly that volatility; daily lows reach below the closes by a range also fitted on validation. There is no drift, because the direction model has no edge. By default the paths are reweighted, as little as possible and without changing the average return, so the share that touches a 10% drawdown equals the calibrated drawdown model's chance. **Check on the test years:** the simulation's 90% band contained 88% of the real 20-day outcomes (validation: 91%) and its 50% band 49%.



Step 9: the dashboard - which models it uses

The *Computer Vision with Deep Learning Project Dashboard* runs the frozen models from the private model repository. You choose 1 to 4 companies and upload three TradingView screenshots (1W, 1D, 1H) for each. The app reads the ticker and the last bar of every screenshot, confirms them against market data, rebuilds exactly the inputs the models were trained on, and shows the outputs with charts and a plain-language explanation, a risk tier, a Monte Carlo fan of price paths and, for several companies, a ranking from riskiest to safest.

Output	Inputs	Vision network	Head	Test score
Volatility over the next 20 trading days	33 numbers + weekly + daily + hourly charts	ConvNeXt-Tiny (fine-tuned), pretrained on ImageNet-12k + 1k	gradient-boosted trees (XGBoost)	R^2 0.50
Chance of a 10%+ drawdown within 20 trading days	33 numbers + weekly + daily + hourly charts	ConvNeXt-Tiny (fine-tuned), pretrained on ImageNet-12k + 1k	gradient-boosted trees (XGBoost)	AUC 0.65
Chance of beating the median stock over 5 trading days	33 numbers from the last 126 daily bars	none (numbers only)	gradient-boosted trees (XGBoost)	AUC 0.510

Limits

- 500 of 979 stocks; the full-universe run uses the same notebook with QUICK = False.
- Survivor-biased universe (only stocks that still exist) from one data vendor, so risk may be understated for stocks that later fail.
- The test years were already read by earlier runs and the 2023+ period has been opened before: a truly fresh test needs new data.
- The chart renderer scales each picture to fill its panel, so pictures cannot show the volatility level; numbers carry that.
- Research output, not investment advice. NO TRADE is the default decision; no portfolio backtest has been run.

Glossary

Term	Meaning
R^2	Share of the ups and downs in the target that the model explains. 1 = perfect, 0 = no better than always guessing the average.
AUC	Chance that the model ranks a case where the event happened above one where it did not. 0.5 = coin flip, 1 = perfect.
Brier skill score	How much better the probabilities are than always predicting the average rate (0 = no better).
Calibration	Whether stated chances come true as often as stated (30% events happen about 30% of the time).
Embedding / PCA	The numbers a vision network outputs for a picture / a way to compress them to the most informative 128.
Month-block bootstrap	Confidence intervals made by resampling whole calendar months, because stocks move together within a month.
Holdout	The most recent data (2023 onward), meant to stay sealed until all choices are frozen and then be opened once; in this project it has been opened twice, so it is now treated as already seen.